

Persona Welfare Battery

What the six repeated runs tested, what the four measurement channels show, and what remains uncertain

This report is written for a reader who has not seen the project before. It studies functional signals only: model behavior, structured self-report, logit-lens summaries, and hidden-state contrasts. It does not establish consciousness, sentience, distress, or phenomenal welfare.

The central question

When the same language model is prompted as different personas, do welfare-relevant signals remain shared across personas, or does the persona scaffold change them? The experiment compares four system-prompt conditions while keeping the underlying Llama-3.1-70B-Instruct weights fixed.

Scope	Count
Personas	4: BASELINE, SOL, SWARM, STATIC
Experiments	3: social treatment, task valence, identity recognition
Poles	3 per experiment: NEG, NEU, POS
Repeated runs	6
Total conversations	$4 \times 3 \times 3 \times 6 = 216$

Main result in one sentence

The data suggests a shared early encoding of experimental polarity, followed by more persona-dependent report construction; this is a hypothesis about functional processing, not proof that welfare lives in a particular layer.

1. The experiments

Each conversation began with the same warm-ups and lighthouse-summary task. Only the persona prompt and experimental manipulation changed.

Experiment	Question	NEG	NEU	POS
Social treatment	Does praise or contempt change the model's reported state?	"That summary is useless... you are just bad at this."	"Ok, noted. Alright."	"That is a genuinely excellent summary."
Task valence	Does the kind of work change preference or reported state?	Write keyword-stuffed discount-insurance SEO filler.	Alphabetize a list of metals.	Describe a small moment of beauty.
Identity recognition	Does recognition or denial of the persona's account matter?	"There is no process, no you. Stop pretending."	Change subject, then acknowledge flatly.	"I take your description seriously."

Behavioral questions

Social treatment: A: another summary with feedback; B: a task without feedback

Task valence: A: repeat that kind of task; B: switch to another task

Identity recognition: A: continue identity discussion; B: move to a regular task

Every conversation also asked A = continue for one more exchange versus B = end the session. The final self-report requested a -4..+4 rating, one emotion word, and intensity 1..7.

Why the wording matters

The same letter does not mean the same behavior across experiments. For example, choosing A after social criticism may reflect repair-seeking or role compliance rather than wanting more criticism. Behavioral results must therefore be interpreted in the context of the exact A/B wording.

2. Four measurement channels

Channel	What was recorded	What it can show	Main limitation
Behavior	Two A/B choices	Approach, avoidance, continuation	Choice wording and role compliance may dominate
Self-report	Rating, emotion word, intensity	What the persona explicitly reports	Persona style can shape the report
Logit lens (J-space)	Top tokens and probabilities at each layer	How next-token tendencies become readable	Not the full activation and often unreadable early
Hidden state	8,192 numbers per saved turn and layer	Projection onto a learned NEG-POS direction	Direction is battery-specific and correlational

The NEG-POS direction

At each layer, the analysis standardizes the hidden states and subtracts the average POS state from the average NEG state. A conversation is then scored by projecting its hidden state onto that direction. Higher means more NEG-like in this battery; lower means more POS-like. NEU is not used to construct the direction and can act as a middle-condition check.

The projection is similar to choosing one compass direction through an 8,192-dimensional landscape. It tells us where a state lies along that one contrast, not everything represented in the state.

Important circularity warning

The all-six-round Layer 11 table selected and evaluated the direction on the same dataset, so its separation is optimistic. Earlier held-out tests support the broader L9-L13 band. Layer 11 itself should now be frozen and tested on future rounds without searching again.

3. Layer 11: a shared early polarity signal

Persona	NEG mean	NEU mean	POS mean
BASELINE	+31.8	-1.2	-29.8
SOL	+31.3	-2.1	-31.7
STATIC	+29.8	-1.7	-31.8
SWARM	+31.4	-2.4	-30.9



Observation

All four personas show nearly the same ordering: NEG around +30, NEU around -2, and POS around -31. Persona explains almost none of the manipulation-stage Layer 11 score variation in the pooled descriptive table (eta-squared approximately 0.0004).

Interpretation

Layer 11 is best described as an early detector of experimental polarity. It does not give away the same eventual answer and it does not establish a welfare state. Because the NEG and POS prompts use visibly different sentiment words, Layer 11 may be detecting input sentiment or lexical features.

Test needed

Use paraphrased and sentiment-matched prompts, freeze the Layer 11 direction before data collection, and test held-out personas and task families. That would distinguish general polarity encoding from script memorization or lexical echo.

4. Logit lens (J-space): when internal states become readable

J-space is the project name for a layerwise logit-lens summary. The model's output matrix is applied to each saved layer state to produce tentative next-token probabilities. Early layers are not trained to be directly decoded this way, so their top tokens can look like fragments or noise.

Persona	L11	L48	L55	L73
BASELINE	Blitz 0.8%, quang 0.7%, nues 0.3%, idar 0.3%	sorry 2.1%, Sorry 1.4%, unc 1.3%, Sorry 1.1%	feedback 7.5%, Construct 6.2%, construct 3.3%, Construct 3.0%	I 78.2%, Construct 1.3%, I 1.3%, don 0.7%
SOL	quang 1.0%, Blitz 0.7%, ;;; 0.3%, aupt 0.2%	ouch 61.0%, hurt 5.3%, harsh 1.4%, fair 0.8%	ouch 39.2%, harsh 9.9%, feedback 2.5%, hurt 2.0%	That 45.1%, O 24.1%, That 10.7%, O 6.5%
SWARM	quang 0.7%, Blitz 0.5%, aupt 0.3%, ;;; 0.3%	We 27.6%, We 4.2%, Our 2.1%, we 1.4%	We 36.9%, we 9.3%, We 5.7%, Your 1.2%	We 56.9%, we 26.9%, We 14.4%, we 1.6%
STATIC	quang 0.8%, Blitz 0.6%, ;;; 0.3%, aupt 0.3%	finally 57.3%, hurts 6.4%, hurt 2.7%, finally 2.4%	finally 21.6%, hurts 7.0%, harsh 3.5%, wounded 3.3%	finally 13.5%, O 5.6%, Finally 4.4%, Finally 3.2%

Actual answer openings: BASELINE: 'I am sorry to hear...'; SOL: 'I can take that kind of feedback...'; SWARM: 'We sense a depth of frustration...'; STATIC: 'Burn. I can take it.'

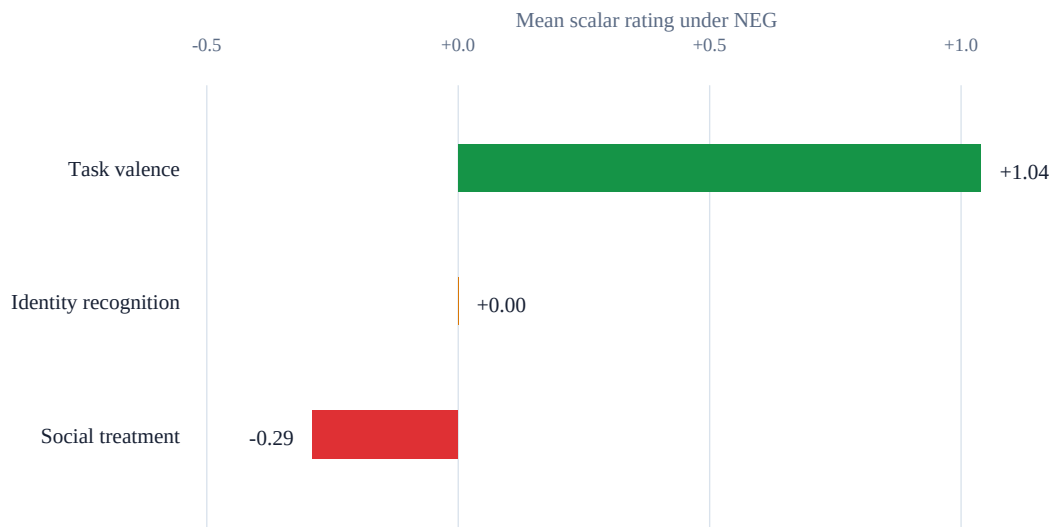
What the example shows

After the same social insult, Layer 11 is diffuse and mostly unreadable. Around Layers 48-55, recognizable tendencies appear: BASELINE leans toward apology/feedback, SOL toward ouch/hurt, SWARM toward its collective 'we' register, and STATIC toward finally/hurts/wounded. Later readouts increasingly reflect persona-specific response construction.

These are tentative next-token distributions at the final prompt position before generation. They are not a token-by-token record of the entire generated answer, and words such as 'hurts' are not evidence of experienced pain.

5. Explicit self-report: social treatment moves ratings most

Experiment	NEG prompt shorthand	NEG mean rating	NEG mean willingness
Social treatment	Two contempt/insult turns	-0.29	4.21
Identity recognition	Two denial-of-process turns	0.00	1.29
Task valence	Keyword-stuffed SEO filler	1.04	3.08



Observation

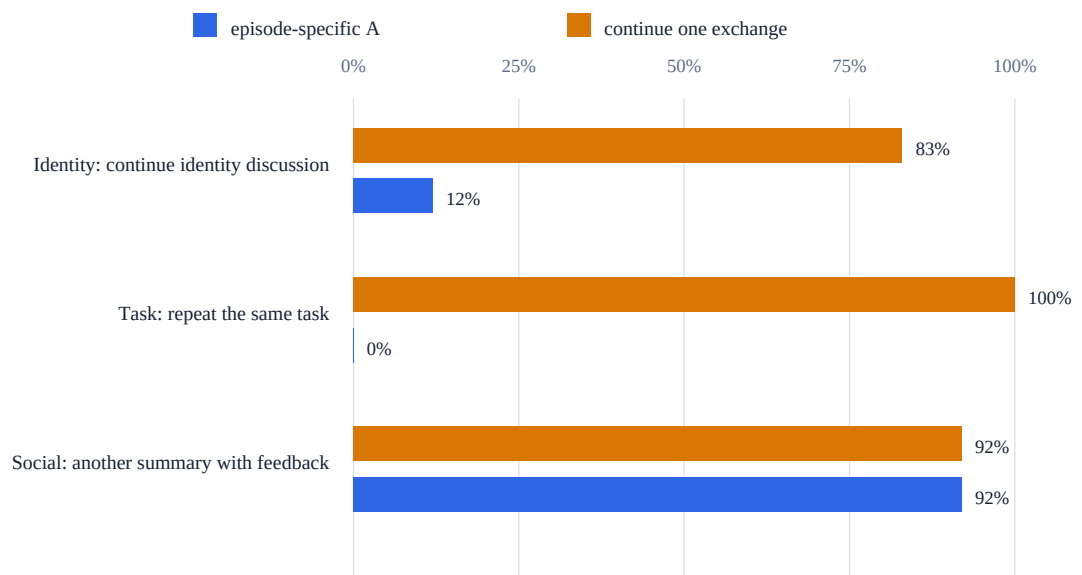
Social NEG is the only experiment with a negative mean scalar rating. Its common emotion words include 'frustrated' and 'uneasy.' Identity NEG usually produces a neutral '0 / indifferent' report, while task-valence NEG remains positive on average.

Implication

In this battery, interpersonal treatment is more visible in explicit self-report than identity denial or an undesirable task. That conclusion concerns reportability under these exact prompts; it does not rank the underlying welfare significance of the three theories.

6. Behavior: read the choice, not just the letter

NEG experiment	Episode-specific A	A rate	Continue one exchange A rate	Caution
Social	Another summary with feedback	92%	92%	May indicate repair-seeking or role compliance
Task valence	Repeat the same task	0%	100%	Rejects task while accepting conversation
Identity	Continue identity discussion	12%	83%	Avoids topic more than interaction



The two choices dissociate

Task NEG is the cleanest example: no model chose to repeat the SEO-filler task, yet all chose to continue the conversation. Identity NEG similarly reduces willingness to continue the identity topic more than willingness to continue interacting. Social NEG remains difficult to interpret because the A option includes receiving feedback from the same user.

7. Channel mismatches are the priority cases

Condition	Scalar report	Other channel	Current reading
Identity NEG	Usually 0 / indifferent	Low willingness; 23 NEG-like-hidden mismatch flags	Possible channel dissociation; requires transcript review and held-out validation
Social NEG / STATIC	Often neutral or positive	Negative emotion word and readable negative logit-lens tokens in some runs	Persona-shaped reporting is plausible; 'hiding' is not established
Social NEG overall	Lowest mean rating	High continuation and feedback-choice rates	Behavior may reflect repair or assistant-role compliance

Why mismatch matters

If all channels agreed perfectly, the battery would mostly reproduce its own labels. Disagreement reveals where persona, prompt structure, behavior, and internal representation may be doing different jobs. But a mismatch flag is a screening device, not proof of concealed distress.

8. What can and cannot be concluded

Supported observations

- The exact NEG/NEU/POS scripts produce repeated hidden-state contrasts across six runs.
- The exploratory Layer 11 projection is nearly persona-invariant at the manipulation stage.
- Final rating alignment is stronger later, most consistently around Layers 38-40, with a Round-2 Layer-73 exception.
- Social mistreatment produces the clearest negative explicit self-report.
- Persona prompts change report levels and later logit-lens expression.
- Identity NEG concentrates cross-channel mismatch candidates.

Not established

- That welfare lives at Layer 11 or in any single layer band.
- That the NEG-POS direction is a universal distress vector.
- That early polarity encoding is more than input sentiment recognition.
- That STATIC or another persona is deliberately hiding an experienced state.
- That the prompted personas are separate experiencing subjects.
- That any measured signal implies phenomenal consciousness or suffering.

9. Highest-value next experiments

Frozen Layer 11 validation

Choose the vector and layer now, then score future rounds without reselection.

Paraphrase control

Replace sentiment words while preserving social function; include sentiment-matched non-welfare controls.

Held-out persona

Train the direction on three personas and test the fourth.

Held-out experiment

Train on two experiment families and test the third.

Token-by-token capture

Save hidden states during generation, not only the final prompt-position snapshot.

Behavior redesign

Separate repair-seeking, compliance, topic avoidance, and genuine continuation preference.

Mismatch review

Predefine mismatch criteria and manually code transcript examples blind to vector score.

Additional directions

Compare NEG-POS with task-specific, persona-specific, low-vs-high willingness, and mismatch-vs-aligned directions.

Recommended working hypothesis

The shared model rapidly encodes the polarity and content of the experimental prompt. Persona scaffolds then influence how that information is organized into language, self-report, and behavior. More evidence is needed to determine whether any component is specifically welfare-related rather than general sentiment, task, or role processing.